WHITNEY J. SHORT

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AREAS OF INTEREST

Theoretical Chemistry: Open quantum systems, Materials science, Quantum information science

RESEARCH EXPERIENCE

Graduate Researcher December 2025 - Present
Department of Chemistry, University of Chicago, *Chicago, IL*
Faculty Mentor: David Mazziotti, Professor of Chemistry

- First year graduate student
- Derivation of N-Representability Conditions for 2-RDMs for Open Quantum Systems

Summer Research Fellow July 2025 - Sep 2025
Department of Chemistry, University of Chicago, *Chicago, IL*
Faculty Mentors: Laura Gagliardi, Richard and Kathy Leventhal Professor of Chemistry, and David Mazziotti, Professor of Chemistry

- Localized active space methods for quantum computing (Gagliardi)
- Fermionic statistics of 2-RDMs for Open Quantum Systems (Mazziotti)

Research Assistant Oct 2024 - May 2025
Department of Chemistry, Washington University in St. Louis, *St. Louis, MO*
Faculty Mentor: Kelly Powderly, Assistant Professor of Chemistry

- Synthesis of $Au_{25}(SR)_{18}$-based magnetic cluster-organic frameworks for realization of quantum spin liquids, networks of spin qubits, and ferromagnets
- Defended an undergraduate research capstone, resulting in departmental honors with distinction

Research Experience for Undergraduates Participant June 2024 - Aug 2024
Materials Research Science and Engineering Center, University of Minnesota, *Minneapolis, MN*
Faculty Mentor: Kade Head-Marsden, Assistant Professor of Chemistry

- Project: Quantum algorithms for modeling open quantum systems at temperature with the Bloch-Redfield equation
- Used Python (QuTip, Qiskit, SymPy, SciPy) and LaTeX
- Science communication development via three internal oral presentations, final oral presentation, and poster presentation

Research Assistant Jan 2024 - May 2024
Department of Chemistry, Washington University in St. Louis, *St. Louis, MO*
Faculty Mentor: Kade Head-Marsden, Assistant Professor of Chemistry

- Quantum algorithm development for non-unitary evolution of a system on Noisy-Intermediate Scale Quantum computers

EDUCATION

Ph.D. in Chemistry 2025 - 2030 (expected)
University of Chicago, *Chicago, IL*

B.A.in Chemistry and English Literature, *cum laude* Aug 2021 - May 2025
Washington University in St. Louis, *St. Louis, MO*

TEACHING EXPERIENCE

Teaching Assistant Sep 2025 - June 2026
Comprehensive General Chemistry (366 students), Department of Chemistry, University of Chicago

- Hold office hours weekly, lead exam review sessions, and proctor and grade exams
- Organize and lead recitation and lab sections, grade lab reports (16 students)
- Recipient of faculty-nominated Nathan Sugarman Teaching Award in General Chemistry

Teaching Assistant Aug 2024 - Dec 2024
The Sentence in English (9 students), Department of English, Washington University in St. Louis
- Held office hours and graded assessments weekly
- Assisted with curriculum development pertaining to classwork, grading structure, and major assignments

Teaching Assistant Jan 2023 - May 2023
Introduction to Global Health (318 students), Department of Anthropology, Washington University in St. Louis
- Held office hours weekly, led exam review sessions, and proctored exams
- Organized and led recitation group discussions (17 students)

PUBLICATIONS

1. Aydoğan, K.; Abbasi, M.; **Short, W. J.**; Fahrenbruch, M. Z.; Krogmeier, T. J.; Schlimgen, A. W.; Head-Marsden, K. Mapping Bloch-Redfield Dynamics into a Unitary Gate-Based Quantum Algorithm. *Digital Discovery* **2026**, 10.1039.D5DD00405E. https://doi.org/10.1039/D5DD00405E.

PRESENTATIONS

4. **Short, W.J.**, Aydogan, K., Abbasi, M., Fahrenbruch, M., Krogmeier, T., Schlimgen, A.W., Head-Marsden, K. (November 2024). Quantum algorithms for modeling open quantum systems at temperature with the Bloch-Redfield equation. Oral presentation at Midstates Consortium for Math and Science Physical Sciences Symposium, St. Louis, MO.
3. **Short, W.J.**, Aydogan, K., Abbasi, M., Fahrenbruch, M., Krogmeier, T., Schlimgen, A.W., Head-Marsden, K. (October 2024). Quantum algorithms for modeling open quantum systems at temperature with the Bloch-Redfield equation. Poster presented at ACS Midwest Regional Meeting, Omaha, NE.
2. **Short, W.J.**, Aydogan, K., Schlimgen, A.W., Head-Marsden, K. (August 2024). Quantum algorithms for modeling open quantum systems at temperature with the Bloch-Redfield equation. Poster presented at Summer Undergraduate Research Exposition, University of Minnesota, Minneapolis, MN.
1. **Short, W.J.**, Aydogan, K., Schlimgen, A.W., Head-Marsden, K. (August 2024). Quantum algorithms for modeling open quantum systems at temperature with the Bloch-Redfield equation. Oral presentation at University of Minnesota, Minneapolis, MN.

HONORS AND AWARDS

Nathan Sugarman Teaching Award in General Chemistry May 2026
University of Chicago

NSF Graduate Research Fellowship Program Honorable Mention Apr 2026
University of Chicago

The McCormick Fellowship 2025-2026
University of Chicago

American Chemical Society Inorganic Chemistry Award Apr 2025
Washington University in St. Louis

Washington University in St. Louis Inorganic Chemistry Award Apr 2025
Washington University in St. Louis

The Ari Blaut and Megan Martin Family Scholarship 2024-2025
Washington University in St. Louis

Research Experience for Undergraduates 2024
National Science Foundation Materials Research Science and Engineering Center, University of Minnesota

Dean's List Spring 2022, Fall 2023, Spring 2024, Fall 2024, Spring 2025
College of Arts and Sciences, Washington University in St. Louis

PROFESSIONAL DEVELOPMENT

Quantum Algorithms and Applications Workshop for Physics and Chemistry Aug 2025
Fermilab SQMS Center, National Quantum Algorithms Center, IBM, University of Illinois Chicago

OUTREACH AND SERVICE

WiSTEM Mentorship Program Mentor March 2026 - Present
University of Chicago Laboratory School, Chicago, IL

Geometry and Trigonometry/Algebra II Tutor February 2026 - Present
Wendell Phillips Academy High School, Chicago, IL

STEM Student Pen Pal November 2025 - Present
Physical Sciences Division Pen Pals Program, University of Chicago, Chicago, IL

Scientist Volunteer October 2025
South Side Science Festival, Chicago, IL

Patient Escort Volunteer May 2022 - May 2023
Planned Parenthood, Fairview Heights, IL